

Capstone Project: Forecaster

Integrity Risk Assessment Agent

Project Overview

The **Integrity Risk Assessment Agent** is a reasoning driven component within a unified integrity platform designed to evaluate whether a company is safe to partner with. Organizations today face a major challenge: risk relevant information is scattered across many sources. Financial stability indicators, regulatory actions, sanctions, ESG controversies, reputational signals, and operational disruptions all live in different systems, formats, and data feeds. Human analysts must manually gather and interpret this information, which is slow, inconsistent, and prone to oversight.

The agent addresses this problem by acting as an intelligent intermediary between the user and a complex ecosystem of data sources. The intended users are **risk and compliance professionals, procurement teams, legal departments, and business stakeholders** who need fast, evidence-based assessments of third-party risk. These users typically ask natural language questions such as “Is Company X safe to partner with?” or “Are there any recent red flags I should know about?” The agent provides a consolidated, real-time view of risk that supports safer and more informed decisions.

Why a stand-alone language model or simple prompting is insufficient

A stand-alone language model cannot solve this problem for several reasons.

First, **LLMs lack real time awareness**. They cannot independently fetch live news, updated sanctions lists, market movements, or newly filed regulatory documents. Integrity assessments require current, verifiable information, not static knowledge from an LLM’s training data.

Second, **LLMs hallucinate when asked factual or risk critical questions**. If prompted directly with “Is Company X safe to partner with?”, a stand-alone model will generate an answer based on patterns rather than evidence. In a compliance or procurement context, this is unacceptable.

Third, **LLMs cannot run forecasting or risk scoring models**. Integrity analysis often requires time series forecasting, anomaly detection, or structured scoring frameworks. These require external tools and models that an LLM alone cannot execute.

Fourth, **LLMs have no persistent memory**. They cannot maintain historical risk profiles, track changes over time, or compare current signals with past assessments.

Finally, **LLMs cannot orchestrate multi step workflows**. A simple prompt cannot coordinate the sequence of actions required: identifying the company, calling multiple APIs, retrieving documents, running models, reconciling conflicting signals, and synthesizing a final answer.

For these reasons, a stand-alone LLM is insufficient. The problem demands an agent that can interact with tools, data sources, and feedback mechanisms.

Environment

The agent operates in an environment that includes:

- **Documents:** news articles, regulatory filings, ESG reports, historical assessments
- **Structured data:** market data, sanctions lists, litigation records, environmental data
- **Tools:** API connectors, retrieval/RAG systems, forecasting models, databases, vector stores
- **Users:** risk/compliance officers, procurement managers, legal teams, business stakeholders

The agent must navigate this environment autonomously to gather evidence and produce grounded assessments.

Actions the Agent must take

To answer, “Is Company X safe to partner with?”, the agent must:

1. Interpret the user’s intent and identify relevant risk dimensions.
2. Plan an information gathering strategy across multiple data sources.
3. Retrieve real time data and relevant documents.
4. Analyze and structure the information (summaries, sentiment, event extraction).
5. Run forecasting and risk models.
6. Resolve conflicting signals and assess confidence.
7. Synthesize a final, justified answer.

These are goal directed actions, not simple text generation

How feedback guides the agent

Feedback improves the agent over time:

- User feedback adjusts reasoning depth, explanation style, and retrieval behavior.
- Environmental feedback (API failures, missing data) triggers retries or alternative strategies.

- Self-reflection helps the agent verify whether its answer is supported by evidence.
- Long-term feedback from historical outcomes recalibrates risk thresholds and model weights.

Feedback transforms the agent from a static pipeline into an adaptive, self-correcting system.

Note, *I used Microsoft Copilot as a writing support tool to help structure and refine the language in this project description. All ideas, analysis, and final content decisions are my own.*